



🔒 Intermediate

Building Your First Streaming Platform

Assemble a working end-to-end streaming stack: ingest with RTMP or SRT, transcode into an ABR ladder, package for HLS/DASH, and deliver through a CDN. Covers origin servers, caching strategy, and the tradeoffs of managed versus self-hosted infrastructure.

🕒 15 - 20 hours

🏗️ Infrastructure & Delivery

📋 6 steps

⚠️ Please log in to start this journey and track your progress.

Learning Path

1

Planning Your Streaming Architecture

Sketch the building blocks of a streaming platform: ingest, transcoder, packager, origin, CDN and player. Decide on live vs VOD scope.

📖 [CDN Transcode Sample](#)



The CDN Transcode Sample, powered by Open Visual Cloud, demonstrates how to integrate media transcoding and streaming services optimized for Intel® Xeon® processors. It utilizes an FFmpeg-based media transcoding stack and an NGINX-based streaming and web hosting stack. This sample is beneficial for developers aiming to build efficient and scalable video streaming solutions with CDN integration.

📖 [MediaPlatform Video Delivery Services](#)



MediaPlatform offers enterprise video delivery solutions that optimize live and on-demand video streaming across corporate networks. Their services include eCDN (Enterprise Content Delivery Network) for efficient video caching and delivery, Smartpath for rules-based distribution, and integration with various peering and WAN optimization technologies. These solutions ensure high-quality video experiences with minimal network impact, making them valuable for organizations looking to enhance their internal video communications.

📖 [Eyevinn Live Encoding](#)



Eyevinn Live Encoding is an open-source live encoder based on ffmpeg and Shaka packager. It supports live transcoding to HLS and optionally MPEG-DASH, providing an

origin for CDN shield to pull streams as well as push to CDN origin. It's a valuable tool for setting up live streaming infrastructures.

2 Setting Up Streaming Servers

Stand up an origin/ingest server (nginx-rtmp, MediaMTX, or similar) and verify you can publish and pull a test stream end-to-end.

CDN Video Uploader and Transcoder



A tool designed for transcoding and uploading videos to a CDN network for HLS video streaming. It supports various input formats and resolutions, utilizes hardware-accelerated video transcoding, and automates the upload process to the target CDN.

Apache Traffic Server



An open source HTTP/1.1 and HTTP/2 caching proxy server that can be used as an edge cache in video streaming content delivery networks (CDNs) to improve scalability.

Netflix Titus (Container orchestration for video)



Netflix's open source container management platform that runs video processing workloads among others. While not video-specific, Titus is key to how Netflix deploys and scales encoding and other batch jobs in a containerized fashion, ensuring efficient resource usage.

3 Building the Encoding Pipeline

Configure a multi-bitrate ladder, segment for HLS/DASH and learn the trade-offs between CPU cost, latency and output quality.

Apache Traffic Control



An open source CDN control plane originally contributed by Comcast. Apache Traffic Control can create a CDN with edge cache servers (like Apache Traffic Server) and provides tooling to manage content delivery, making it useful for building an edge caching infrastructure for video.

Creating a secure video-on-demand (VOD) platform using AWS



Authored by Chirag Oswal, Solution Architect, AWS, and Vikas Tiwari, Solution Architect Manager, AWS Video has become the primary means of Information sharing and learning. Customers are investing in innovative solutions to tap into the e-learning and video space. Video content is their IP and needs to be protected and securely delivered. Online video is a [...]

lhls-simple-live-platform



You can build your own live platform just wiring up some open source tools, this is a demo video of [HTTPS://GitHub.com/jordicenzano/hls-simple-live-platform](https://github.com/jordicenzano/hls-simple-live-platform).

4 CDN Integration & Caching

Front your origin with a CDN, tune cache headers for segments vs manifests, and measure delivery performance from multiple regions.

[OTT Content Delivery– CDN Alternatives](#)

Article exploring CDN alternatives for OTT content delivery. Covers infrastructure options for video streaming.

[Edge Computing with Fastly CDN and Varnish VCL for Authenticated Requests - Endertech](#)

A brief explanation of how to use Fastly CDN and Varnish with authenticated requests to offload static file serving from the origin and edge caching.

[Understanding Server-Side Dynamic Ad Insertion](#)

In this post we're explaining the principles behind Server-Side Dynamic Ad Insertion technology.

5 Player Implementation

Embed an adaptive player, wire up basic analytics events and confirm graceful degradation on poor networks.

[OTT Content Delivery– Multi CDN](#)

Eyevinn article explaining multi-CDN strategies and architectures for OTT video content distribution.

[Google Media Transcoder](#)

Google Media Transcoder is a cloud-based service that provides scalable and efficient video transcoding capabilities. It supports a wide range of input and output formats, enabling developers to deliver high-quality video content across various devices and platforms.

[OblivCDN: A Practical Privacy-preserving CDN with Oblivious Content Access](#)

OblivCDN is a privacy-preserving system designed for seamless integration with existing real-world Internet-CDN infrastructure. It adapts Range ORAM primitives to optimize memory and disk seeks when accessing contiguous blocks of CDN content, preserving

both content confidentiality and user access pattern hiding features. OblivCDN eliminates the need for trusted hardware on edge servers, enhancing compatibility with networks containing legacy devices. In real-world streaming evaluations, it demonstrated remarkable performance, downloading a 256 MB video in just 5.6 seconds.

6 Testing, Monitoring and Optimization

Load-test the pipeline, set up QoE monitoring (rebuffer ratio, startup time) and iterate on bottlenecks before launch.

[hlsjs-p2p-engine](#)



hlsjs-p2p-engine enhances video streaming scalability by implementing WebRTC data channels to create a peer-to-peer (P2P) network overlaying existing CDNs or origin servers. Compatible with hls.js, it supports both live and VOD streams over the HLS protocol and integrates seamlessly with popular HTML5 players like video.js, JWPlayer, and Flowplayer. This approach reduces server load and bandwidth costs while improving streaming performance, offering developers an innovative solution for efficient and scalable video delivery.

[Example Project: Video Streaming Platform](#)



An example of a Video on Demand (VOD) streaming platform built with open web services in Eyevinn Open Source Cloud. It includes a VOD preparation pipeline, orchestrator, database, and a simple web application.

[Qencode Cloud Video Services](#)



Qencode offers a suite of cloud-based video services, including video transcoding, live streaming, media storage, and content delivery. Their APIs allow developers to transcode videos into various formats and codecs, launch high-quality live streams globally, and manage media storage efficiently. The platform also provides a customizable HTML5 video player and features like per-title encoding to optimize file sizes using AI-powered video optimization.